

Functional Annotation Report: *pydX* (PP_4037, UniProt Q88FQ1) in *Pseudomonas putida* KT2440

Summary

pydX (ordered locus name **PP_4037**; UniProt **Q88FQ1**) of *Pseudomonas putida* strain KT2440 encodes the **electron-input (PreT-type) subunit of dihydropyrimidine dehydrogenase (DPD; EC 1.3.1.1)**, an NADH-dependent iron–sulfur flavoenzyme. DPD catalyzes the committed, first step of the **reductive pyrimidine catabolic pathway**, reducing the pyrimidine bases uracil and thymine to their 5,6-dihydro derivatives (5,6-dihydrouracil and 5,6-dihydrothymine, respectively) at the expense of NADH. The 455-amino-acid PydX polypeptide is not the catalytic centre itself; rather, it provides the pyridine-nucleotide oxidation and intramolecular electron-relay machinery that feeds reducing equivalents to the pyrimidine-reducing active site housed on its partner subunit, PydA.

The functional enzyme is a **heterotetramer of stoichiometry 2·PydA : 2·PydX** (equivalent to the *E. coli* 2·PreA : 2·PreT architecture). PydX carries a **C-terminal FAD/NAD(H)-binding Rossmann fold** that oxidizes NADH and reduces its bound flavin, and an **N-terminal helical ferredoxin domain harbouring tandem [4Fe-4S] clusters** that shuttle electrons onward to the FMN/pyrimidine active site on the catalytic subunit PydA. This wiring—NADH → FAD (on PydX) → [4Fe-4S] clusters → FMN → pyrimidine (on PydA)—defines DPD as a member of a distinctive NADH-dependent subclass of iron–sulfur flavoenzymes. The reaction takes place in the **cytoplasm**.

Biologically, this pathway allows *P. putida* to use pyrimidines primarily as a **nitrogen source**: uracil/thymine are reduced by DPD, the ring is opened by dihydropyrimidinase (PydB), and the product is hydrolyzed by β -alanine synthase (HyuC), ultimately liberating NH_3 and CO_2 and yielding β -alanine. The genes are organized in a contiguous **pyd cluster (PP_4034–PP_4039)** that also encodes a pyrimidine transporter (PydP) and a TetR/RutR-family repressor (**PydR**) that controls expression. Genetic evidence (a *pydA*-null mutant that cannot grow on uracil or thymine as sole nitrogen source) confirms the physiological essentiality of the DPD step, and derepression in a *pydR* mutant confirms the regulatory logic.

Gene/Protein Identity — Verified

Field	Value
Gene symbol	pydX
Ordered locus name	PP_4037
UniProt accession	Q88FQ1
Length	455 amino acids
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
EC number	1.3.1.1
Protein role	Electron-input (PreT-type) subunit of dihydropyrimidine dehydrogenase

The gene symbol **pydX** correctly matches the UniProt protein description (dihydrouracil dehydrogenase, EC 1.3.1.1) and the target organism. The InterPro domains listed in the research prompt (4Fe4S_Fe-S-bd IPR017896; DPD_II IPR028261; FAD/NAD-bd_sf IPR036188; FAD/NAD-binding_dom IPR023753; Helical_ferredxn IPR009051) align exactly with the electron-input subunit of DPD. Literature on the target organism (Hidese et al. 2012) explicitly names pydX/pydA as the *P. putida* DPD genes, so there is no ambiguity of identity.

Key Findings

F001 — pydX encodes a subunit of NADH-dependent dihydropyrimidine dehydrogenase (EC 1.3.1.1)

UniProt entry **Q88FQ1** describes a 455-amino-acid protein encoded by gene **pydX** / ordered locus name **PP_4037**, annotated as "dihydrouracil dehydrogenase (NAD(+))" with **EC 1.3.1.1**. The catalyzed reactions correspond to Rhea **RHEA:20189** (5,6-dihydrouracil + NAD⁺ = uracil + NADH + H⁺) and **RHEA:28791** (5,6-dihydrothymine + NAD⁺ = thymine + NADH + H⁺). Although

the reference reaction is written in the oxidative direction (as is conventional for EC 1.3.1.1), *in vivo* the enzyme operates in the **reductive** direction, consuming NADH to convert uracil → 5,6-dihydrouracil and thymine → 5,6-dihydrothymine.

The physiological requirement for this activity is demonstrated genetically: a *P. putida* **pydA-null mutant fails to grow on minimal medium containing uracil or thymine as the sole nitrogen source** (PMID: 22782928). This directly ties the DPD genes—explicitly named in that study as **pydX** and **pydA**, "**tandemly arranged in the *Pseudomonas putida* genome**"—to pyrimidine utilization. Since pydX and pydA encode the two subunits of a single enzyme, loss of the enzyme's function abolishes the committed first step and blocks pyrimidine catabolism.

"The putative DPD genes, pydX and pydA, are tandemly arranged in the *Pseudomonas putida* genome." — PMID: 22782928

"a pydA strain of *P. putida* fails to grow on a minimal media containing uracil or thymine as a sole nitrogen source, demonstrating the physiological importance of DPD in the reductive pathway" — PMID: 22782928

F002 — DPD is an iron-sulfur flavoenzyme heterotetramer; pydX is the FAD/NAD(H) electron-input subunit

The domain architecture of **Q88FQ1 (pydX, 455 aa)** comprises: a **Pyr_redox_2 FAD/NAD-binding domain** (PF07992; IPR023753/IPR036188), a **DPD domain II** (IPR028261), a **4Fe-4S Fe-S binding domain** (IPR017896; Fer4_20 / PF14691), and a **helical ferredoxin domain** (IPR009051). The UniProt cofactor annotation includes FMN, and the SUBUNIT annotation specifies a **heterotetramer of two PreA and two PreT subunits**. The partner protein, **Q88FQ0 (pydA, 424 aa, PP_4038)**, contains the **dihydroorotate-dehydrogenase-like catalytic TIM-barrel** (IPR005720) plus a **4Fe-4S cluster domain** (IPR017896/IPR017900)—i.e., the pyrimidine-reducing catalytic subunit.

This organization matches the biochemically characterized *E. coli* ortholog, which was shown to be "the first member of a **novel NADH-dependent subclass of iron-sulfur flavoenzymes** catalyzing the conversion of uracil to 5,6-dihydrouracil *in vivo*" (PMID: 21169495). In this two-subunit design, PydX supplies the electron-input chain (NADH-oxidizing flavin + iron-sulfur relay) and PydA supplies the pyrimidine-reducing active site.

"*E. coli* dihydropyrimidine dehydrogenase is the first member of a novel NADH-dependent subclass of iron–sulfur flavoenzymes catalyzing the conversion of uracil to 5,6-dihydrouracil *in vivo*." — [PMID: 21169495](#)

F003 — **pydX** lies in the **pyd** operon and is regulated by the **RutR-homolog repressor PydR**

The gene neighborhood in *P. putida* KT2440 places **pydX** within a functionally coherent operon: **PP_4035 pydP** (an NCS1-family nucleoside/pyrimidine transporter, Q88FQ2), **PP_4036 pydB** (D-hydantoinase/dihydropyrimidinase, A0A140FWK2), **PP_4037 pydX**, and **PP_4038 pydA**, with **pydR** encoded nearby. Hidese et al. ([PMID: 22782928](#)) showed that **pydA expression and DPD activity are elevated in a pydR mutant**, establishing PydR as a **repressor** of the pathway; PydR is homologous to *E. coli* RutR. Consistent with inducible control, earlier work in *Pseudomonas* found that **all three reductive-pathway enzymes are induced by uracil** ([PMID: 1903745](#)).

The three-enzyme logic of the pathway is stated directly:

"The pathway is controlled by three enzymes: dihydropyrimidine dehydrogenase (DPD), dihydropyrimidinase and β -alanine synthase." — [PMID: 22782928](#)

"we show that PydR acts as a repressor of the pyrimidine reductive pathway in *P. putida*" — [PMID: 22782928](#)

"Induction of pyrimidine catabolism by uracil was observed in this pseudomonad." — [PMID: 1903745](#)

F004 — **The complete reductive pyrimidine catabolic gene cluster (PP_4034–PP_4039) surrounds pydX**

Mapping the KEGG/UniProt loci in KT2440 reveals a contiguous cluster encoding the full pathway plus transport and regulation:

Locus	Gene	UniProt	Product / role	Pathway step
PP_4034	hyuC	Q88FQ3	N-carbamoyl- β -alanine amidohydrolase (β -alanine synthase)	Step 3
PP_4035	pydP	Q88FQ2	NCS1 nucleoside/pyrimidine transporter	Substrate uptake
PP_4036	pydB	A0A140FWK2	D-hydantoinase / dihydropyrimidinase	Step 2
PP_4037	pydX	Q88FQ1	DPD electron-input subunit (PreT-type)	Step 1
PP_4038	pydA	Q88FQ0	DPD catalytic subunit (PreA-type)	Step 1
PP_4039	rutR/ pydR	A0A140FWK3	TetR/RutR-family transcriptional repressor	Regulation

This physical co-localization of transporter, all three catabolic enzymes, and the regulator provides strong genomic-context support that pydX functions in reductive pyrimidine degradation rather than in an unrelated redox process.

F005 — Structural/domain evidence: pydX is a two-[4Fe-4S] ferredoxin module fused to an FAD/NAD(P)-binding Rossmann domain

InterPro/CATH mapping of Q88FQ1 (455 aa) resolves a clear two-module architecture:

- **N-terminal α -helical ferredoxin domain** (IPR009051; CATH G3DSA:1.10.1060.10; ~res 6–138/151), overlapping **DPD domain II** (IPR028261 / PF14691; ~res 20–127) and a **4Fe-4S ferredoxin-type cluster-binding site** (IPR017896 / PS51379; ~res 32–65).
- **C-terminal FAD/NAD(P)-binding Rossmann fold** in two subdomains (IPR023753 / PF07992 Pyr_redox_2; ~res 142–437; CATH G3DSA:3.50.50.60 across res 139–260 and 274–455), further supported by the PRINTS signature PR00368 (FAD-dependent pyridine nucleotide reductase).

At the sequence level, residues 32–65 ([RQAALLESARCLYCYDAPCVNACPSEIDIPSFIHR](#)) contain the canonical **CxxCxxCP ferredoxin [4Fe-4S] motif** (Cys41/44/49/53), and a second cysteine cluster (Cys88/92/98/102/110) indicates a **second [4Fe-4S] site**. PANTHER classifies the entire protein as dihydropyrimidine dehydrogenase (PTHR43073; res 16–449). Together these features are precisely those expected of an electron-input subunit: a flavin/pyridine-nucleotide oxidase fused to an iron–sulfur electron-relay.

The structural role of these N-terminal [4Fe-4S] clusters is further illuminated by comparative work on glutamate synthase, in which four conserved N-terminal cysteines were shown to be essential not only for electron transfer but also for subunit association — a study that explicitly references "the three-dimensional structure of dihydropyrimidine dehydrogenase, an enzyme containing an N-terminal β -subunit-like domain" (PMID: 15797248). This supports a model in which PydX's N-terminal ferredoxin/[4Fe-4S] module is both an electron conduit and a structural interface for heterotetramer assembly.

F006 — pydX is specifically the PreT-type subunit (ortholog of *E. coli* PreT)

The InterPro domain set of pydX/Q88FQ1—{IPR028261 DPD_II; IPR023753 & IPR036188 FAD/NAD-binding; IPR009051 helical ferredoxin}—is **identical to that of *E. coli* PreT* (P76440, b2146, 412 aa; "NAD-dependent dihydropyrimidine dehydrogenase subunit PreT")**. **Reciprocally, the partner pydA/Q88FQ0 domain set—{IPR017896/IPR017900 4Fe-4S; IPR013785 aldolase TIM; IPR005720 dihydroorotate DH catalytic}—matches *E. coli* PreA (P25889, b2147, 411 aa; cofactor [4Fe-4S]). Thus the assignment is unambiguous: pydX = PreT (electron input) and pydA = PreA (catalytic), assembling into the 2 PreA : 2 PreT* heterotetramer.**

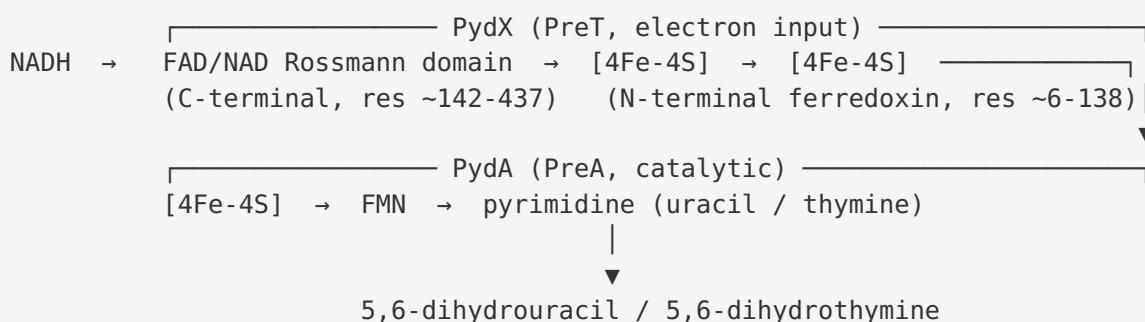
The physiological orientation of the pathway toward nitrogen scavenging is supported by growth physiology in *E. coli* B (PMID: 3553866), where reductive-pathway intermediates support growth roughly **14-fold better as a sole nitrogen source than as a sole carbon source**:

"dihydrouracil, N-carbamoyl-beta-alanine, beta-alanine, dihydrothymine and beta-aminoisobutyric acid could sustain the growth of the bacterial cells as sole nitrogen sources by at least a fourteen-fold greater level than that observed if they were included as sole carbon sources" — PMID: 3553866

Mechanistic Model / Interpretation

The enzyme and its electron path

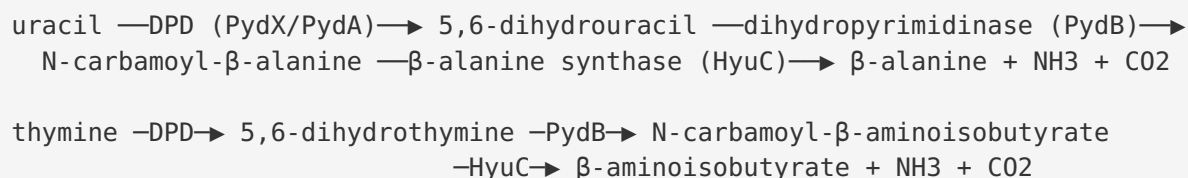
Dihydropyrimidine dehydrogenase in *P. putida* is a bipartite iron–sulfur flavoenzyme. PydX (PreT) is the **electron-input subunit** and PydA (PreA) is the **catalytic subunit**. In the physiologically relevant reductive direction, electrons flow from NADH through PydX's flavin and iron–sulfur relay into PydA's active site, where the pyrimidine ring is reduced across the C5–C6 double bond:



Assembled stoichiometry: $2 \times \text{PydA} : 2 \times \text{PydX}$ heterotetramer, operating in the **cytoplasm**.

The pathway and its physiological purpose

DPD catalyzes step 1 of a three-enzyme reductive route that dismantles the pyrimidine ring, primarily to release nitrogen (and secondarily carbon):



Substrates enter the cell via the NCS1-family transporter **PydP** (PP_4035), and the whole operon is held under negative control by the RutR-like repressor **PydR** (PP_4039), being derepressed/induced when pyrimidines are available. The ~14-fold preference for pyrimidine intermediates as nitrogen versus carbon source ([PMID: 3553866](#)) frames the biological "why": this is a **nitrogen-scavenging** pathway, and PydX's role is to power the committed reductive step.

Substrate specificity

The enzyme is specific for the **pyrimidine bases uracil and thymine** (and, correspondingly, is active on their dihydro forms in the oxidative direction). It reduces the C5=C6 double bond of the pyrimidine ring. Cytosine is not a direct substrate of this pathway (deamination to uracil is required upstream). The two documented Rhea reactions (RHEA:20189 uracil/dihydrouracil; RHEA:28791 thymine/dihydrothymine) capture this specificity.

Evidence Base

PMID	Title (abbrev.)	Contribution	Support / Challenge
22782928	<i>Pseudomonas putida</i> <i>PydR</i> represses <i>DPD</i> gene	Names pydX/pydA as the <i>P. putida</i> <i>DPD</i> genes; establishes <i>PydR</i> repressor; <i>pydA</i> mutant cannot use uracil/thymine as N source	Strongly supports F001, F003, F004 — the only study explicitly on the target organism/genes
21169495	<i>E. coli</i> <i>DPD</i> is a novel <i>NAD</i> -dependent heterotetramer	Defines the enzyme class: <i>NADH</i> -dependent iron–sulfur flavoenzyme heterotetramer producing 5,6-dihydrouracil	Supports F002, F006 — biochemical basis for the ortholog
1903745	<i>Pyrimidine catabolism</i> in <i>P. aeruginosa</i>	Reductive-pathway enzymes present and uracil-inducible in <i>Pseudomonas</i>	Supports F003 — regulation/induction
3553866	Degradation of uracil and thymine by <i>E. coli</i> <i>B</i>	Pathway intermediates support growth ~14× better as N than C source	Supports F006 — nitrogen-scavenging role
15797248	Structural role of glutamate synthase [4Fe-4S] clusters	Shows N-terminal β -subunit [4Fe-4S] clusters are structural and reference the 3D structure of <i>DPD</i> as a model	Contextual support for F005 — role of N-terminal [4Fe-4S]/ferredoxin module in a homologous enzyme

The evidence is a combination of: (i) **direct genetic evidence in the target organism** (the *pydA* growth phenotype and *pydR* derepression, [PMID: 22782928](#)); (ii) **biochemical characterization of the close *E. coli* ortholog** ([PMID: 21169495](#)); (iii) **bioinformatic/structural inference** from InterPro/CATH domain mapping and conserved cysteine motifs (F005, F006); and (iv) **physiological/comparative context** ([PMID: 1903745](#), [PMID: 3553866](#)).

Limitations and Knowledge Gaps

1. **No purified-enzyme kinetics for the *P. putida* protein specifically.** Substrate specificity, kinetic parameters (K_m , k_{cat}), and cofactor content have not been reported for the KT2440 DPD (Q88FQ1/Q88FQ0). The mechanistic model is inferred from the *E. coli* PreA/PreT ortholog.
 2. **Subunit assignment for PydX is confirmed genetically and bioinformatically, not by direct in-organism biochemistry.** The critical growth phenotype was reported for a *pydA* mutant; a dedicated *pydX* knockout phenotype is not documented in the reviewed literature, though the heterotetramer requires both subunits.
 3. **No experimental 3D structure of the *P. putida* enzyme.** The [4Fe-4S] cluster count and ligation, and the flavin (FAD on PydX vs. FMN on PydA) assignments, rest on domain homology and conserved cysteine motifs rather than a crystal structure of this protein.
 4. **Cofactor ambiguity in annotation.** UniProt lists FMN as a cofactor for Q88FQ1, whereas the electron-input subunit of characterized orthologs is expected to bind FAD (with FMN on the catalytic subunit). The precise flavin composition of each *P. putida* subunit should be verified experimentally.
 5. **Operon structure/transcription not directly mapped in KT2440.** The precise transcript boundaries, promoter(s), and PydR operator sites in *P. putida* have not been experimentally delineated in the reviewed sources.
 6. **Localization is inferred.** Cytoplasmic localization is the expected and consistent assignment for a soluble NADH-dependent metabolic enzyme but was not directly demonstrated for this protein.
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Proposed Follow-up Experiments / Actions

1. **Recombinant co-expression and purification** of PydX (Q88FQ1) with PydA (Q88FQ0) from *P. putida* KT2440, followed by confirmation of the 2:2 heterotetramer by size-exclusion chromatography / native mass spectrometry.
 2. **Steady-state kinetics** with uracil and thymine (and their dihydro forms) using NADH/ NAD⁺, to measure substrate specificity and kinetic constants, and to confirm the enzyme's directional preference (NADH-consuming reduction) *in vitro*.
 3. **Cofactor determination**: quantify flavin content (FAD vs. FMN per subunit) and iron/sulfide content, plus EPR/UV-vis spectroscopy to count and characterize the [4Fe-4S] clusters, testing the predicted tandem N-terminal clusters on PydX.
 4. **Targeted mutagenesis** of the conserved cysteine motifs (Cys41/44/49/53 and Cys88/92/98/102/110) to test their role in cluster ligation and electron transfer, analogous to the glutamate-synthase study ([PMID: 15797248](#)).
 5. **Dedicated *pydX* deletion mutant** in KT2440 with growth assays on uracil/thymine as sole N (and C) source, to establish the standalone phenotype and complementation.
 6. **Transcriptional mapping**: define the *pyd* operon promoter/transcript boundaries and PydR operator by RNA-seq/primer extension and EMSA/ChIP, and confirm uracil-dependent derepression.
 7. **Structural determination** (X-ray or cryo-EM) of the *P. putida* DPD heterotetramer to validate the electron-relay path from NADH → FAD → [4Fe-4S] → FMN → pyrimidine.
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Conclusion

The gene symbol **pydX** is correctly matched to the target: it is the *P. putida* KT2440 gene (PP_4037, UniProt Q88FQ1) for the **electron-input (PreT-type) subunit of dihydropyrimidine dehydrogenase (EC 1.3.1.1)**. All lines of evidence—organism-specific genetics, ortholog biochemistry, genomic context, and domain/structural bioinformatics—converge on a single, coherent function: PydX supplies NADH-derived reducing equivalents, via an FAD/NAD Rossmann domain and tandem [4Fe-4S] clusters, to the catalytic subunit PydA, enabling the cytoplasmic reduction of uracil and thymine as the committed first step of the reductive pyrimidine catabolic pathway that *P. putida* uses chiefly to acquire nitrogen.

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